

DHARMALA HRUDAI ADITYA

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Focus: reliability & efficiency of agentic LLM systems fine-tuning, quantization, evaluation & verification, on-device inference.

EDUCATION

Vasavi College of Engineering, Hyderabad

2020 – 2024

B.E. (Honours) in Information Technology, specialization in AI/ML · CGPA 8.96 / 10.0

Research Interests: reliability and evaluation of agentic & semi-agentic LLM systems; efficient on-device fine-tuning under hardware constraints; RL for sequential reasoning.

PUBLICATIONS & RESEARCH

- Dharmala, H. A., et al. “[Visual Question Answering on Remote Sensing Images](#).” Springer (SmartCom 2025). Built a custom U-NET on the RSVQA dataset that surpassed baselines on Hamming distance and F1; integrated as an expert module in an open-source multimodal system, improving “Presence”-type accuracy by **14%** via fine-tuning and controlled prompts.
- Shrivastava, A., Dharmala, H. A. “COVE: A Task-Adaptive Code-Based Ensemble for LLM Output Verification.” Internal technical report, Infosys AI R&D; Built the verification harness and multi-dataset evaluation pipeline; the ensemble improved average text-task verification accuracy by **81.7%** (relative lift) over baseline zero-shot LLM verifiers with **48–96%** token savings, at near-zero marginal cost and sub-millisecond latency.

EXPERIENCE

Specialist Programmer, AI Product Engineering

Sept 2024 – Present

Infosys, Hyderabad, India

- Researched and shipped an on-device agentic invoice system (WPF, Windows ML, ONNX Runtime GenAI). RL-fine-tuned Phi-4 Mini and optimized the **ONNX compute graph**, **INT4 quantization**, and feature maps for efficient deployment to Snapdragon’s Hexagon NPU. Measured against the original cloud-API baseline, on-device inference latency dropped to **566 milliseconds per invoice**, achieving a **3–5× relative speed-up**. [Featured by Microsoft at Ignite 2025](#).
- Researched **LLM output verifiers and RL algorithms to strengthen the core** component of the COVE verification system and rebuilt the SFT data and evaluation pipelines (cleaning, curation, criteria), raising absolute end-to-end classification accuracy from **76% to 91%** on held-out sets across three deployments.
- Built the long-term-memory and tool-orchestration layer of [Infosys Agentic Foundry](#), improving MCP-enablement completion across finance and energy engagements.

Applied Researcher

Apr 2024 – Sept 2024

Genloop, Bangalore, India

- Worked on the knowledge-graph and agentic backend for Genloop’s Text-to-SQL / Data-Analyst platform (Clarification Graph + LLM-observability layer) and ran GRPO-based RL fine-tuning with systematic reward-design ablations. Laid the groundwork for leaderboard performance in [spider 2.0](#)
- Designed **multi-adapter LoRA fine-tuning** pipelines on GCP and built **root-cause-analysis (RCA)** capabilities for data anomalies across conversational-analytics workflows; led the weekly “[Tuesday Papers](#)” research-dissemination program.

AI Labs Intern

Jun 2023 – Mar 2024

Brane Enterprises @ NSL HUB, Hyderabad, India

- Deployed LLMs on NVIDIA Jetson Xavier with Kubernetes + Ray Serve at a **sub-300 ms** latency target; as part of Brane’s [Darwin Center](#) research initiative, built a **1M+** example dataset and fine-tuned V100 models for **biomedical tasks** across 6 datasets, improving task metrics **12–20%** over the pre-fine-tuning baseline.

OPEN SOURCE CONTRIBUTION

[Infosys Agentic Foundry](#) open source (MIT), part of Infosys Topaz

- Core contributor to **Infosys Agentic Foundry**, an open-source agentic-AI platform: worked on agent memory/tool orchestration and **ground-truth evaluation framework** recognized with the Infosys **OSPO CodeCon Star**.
- Built containerized multi-agent services with end-to-end Open Telemetry tracing into Elasticsearch + Grafana and Arize Phoenix for LLM evaluation, run comparison, and regression detection.

AWARDS, HONORS & LEADERSHIP

- IEEE Student Branch Chair:** led the GRSS (GRS29) and CIS (CIS11) chapters at Vasavi College; organized 3 national and 1 international conference [MIGARS] workshop during tenure.
- NPTEL Discipline Star (Computer Science & Engineering):** Recognized for completing over 50 weeks of advanced coursework alongside core degree.
- Infosys Award for Excellence 2025–26:** presented by CEO & MD Salil S. Parekh (department flagship recognition given to only 0.1%).
- Vidyarthi Vigyan Manthan (NCERT / VIBHA):** cleared district and state levels to represent the state at the national camp (IIT Delhi)
- Hackathon commendation:** TSIC Foundation (Govt. of Telangana) × UMass Amherst (DACSS), Infosys × Stanford Digital Economy Lab
- Google DSC Lead:** 1 of 400 nationwide (500+ peers); taught AI/ML sessions across college developer forums. · Amazon ML Summer School.

SKILLS

Languages: Python, Java, JavaScript, SQL, C#, React

AI / ML: LLM fine-tuning (SFT, LoRA, AWQ INT4), RL (GRPO, RLVR), model evaluation & output verification, RAG, computer vision

ML Infra: ONNX Runtime GenAI, Olive, QNN Execution Provider, Ray Serve, FastAPI, data-processing pipelines

Cloud & Agentic: GCP, Azure, Kubernetes, Docker, LangGraph, Google ADK, MCP, ChromaDB, FAISS, OpenTelemetry

CERTIFICATIONS

Microsoft Azure AI Engineer Associate · NVIDIA-Certified Associate: Accelerated Data Science · Google Cloud Professional Cloud Architect · Claude Certified Architect (Anthropic) Data Science & AI, University of Cambridge · NPTEL Linear Algebra